Sample Preparation

After sample received: 5ul of liquid sample was resuspend in 30ul of 100mMTris, pH8.5 with 8M Urea. TCEP and CAM treated for 30mins in dark, then dilute Urea to 2M by Tris and add 5ul of 0.1 ug/ul Trypsin/LysC (Promega # V5071) on each sample at 37 degrees overnight. Adding 7ul of FA to stop reaction. Aliquot 2x 25ul of reaction on two EvoTips for DDA on Flex.

LC-MS Analysis of Peptides

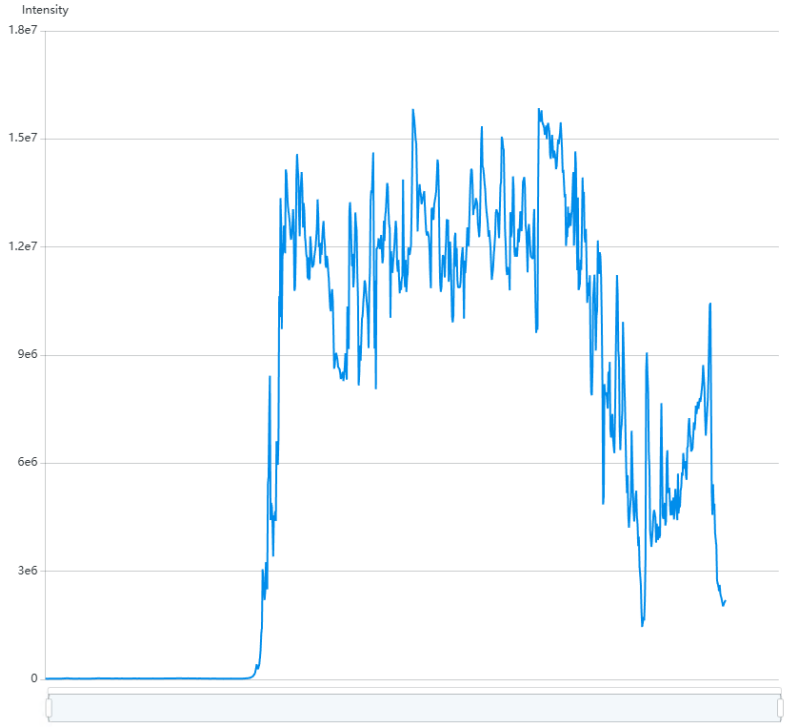
UHPLC

Peptides were resolved using an Evosep One EV1000 UHPLC in Zoom mode connected to an Aurora Elite CSI 15 cm x 75 µm C18 UHPLC column (ionopticks) in turn connected to a CaptiveSpray source. The Whisper Zoom 40SPD chromatography method with a runtime of 32.5 minutes was used with Buffer A (0.1% formic acid in LC-MS grade water) and Buffer B (0.1% formic acid in LC-MS grade acetonitrile).

Mass Spectrometry

DDA mass spectrometry analysis used a timsTOF Flex MALDI 2 instrument run in positive ion mode.

DDA data: Total Ion Chromatogram (TIC) for all samples



Data Analysis

DDA Data Analysis with FragPipe23.1

This database contains *Ecoli.* proteins downloaded from UniProt on 2025 and 381 contaminant proteins as published by Frankenfield et al. in 2022[3]. Also include 1 recombinational sequence that is shared by CPR `Ant-Hox1a Sequence.`

DDA Search method in FragPipe chose LFQ-MBR as Workflow and keep rest as default.

Results

Mass spectrometry Overview

Following the initial searches, we identified the following in each sample from DIA and DDA:

Experiment	#Proteins	#PSMs	#Peptides
AntHOX1a	723	6235	2924

Protein Identification & Quantitation

The **bait protein,** Ant-Hox1a(14.9%) was on the top of the table based on their **Combined Total Spectral Count** values in DDA-Pr. Trypsin and Protease IV are the proteins on the top of contaminants which are added in. Protein contaminants such as keratins or enzymes are listed as well. We use the ratio of Spectral Count from contaminant proteins over total protein PG.Quantity to estimate contamination levels. The contamination level is okay since major contaminants are Trypsin and rLys-C we added in.

	Total Spectral Count	Intensity
Total	6264	74642457
Contaminant	592	7803219
Contaminant %	9.45	10.45
Ant-Hox1a	931	17898370
Ant-Hox1a %	14.86	23.98

The AntHox1a coverage shows as below.

```
M D C Y F P V T N T F N P P A F C A M P S M M Y T Y G M R S Y Y P P A A
L Y P L A S A S L T E R S N G P I M N S Q V Q A D Q N S A I N Q G M E L
R N T A S S G K L Q M S G K G C D S Y V R L L S R N E P A E K E L E R S
D D E C E D S K S E E S G D A P W H Y K Q P F K H R K R M A Y T R I Q L
L E L E K E F H F T R Y L T K E R R T E M A R M L D L T E R Q V K I W F
Q N R R M K W K K D N K P V I P T S R G M R K G C T R
```